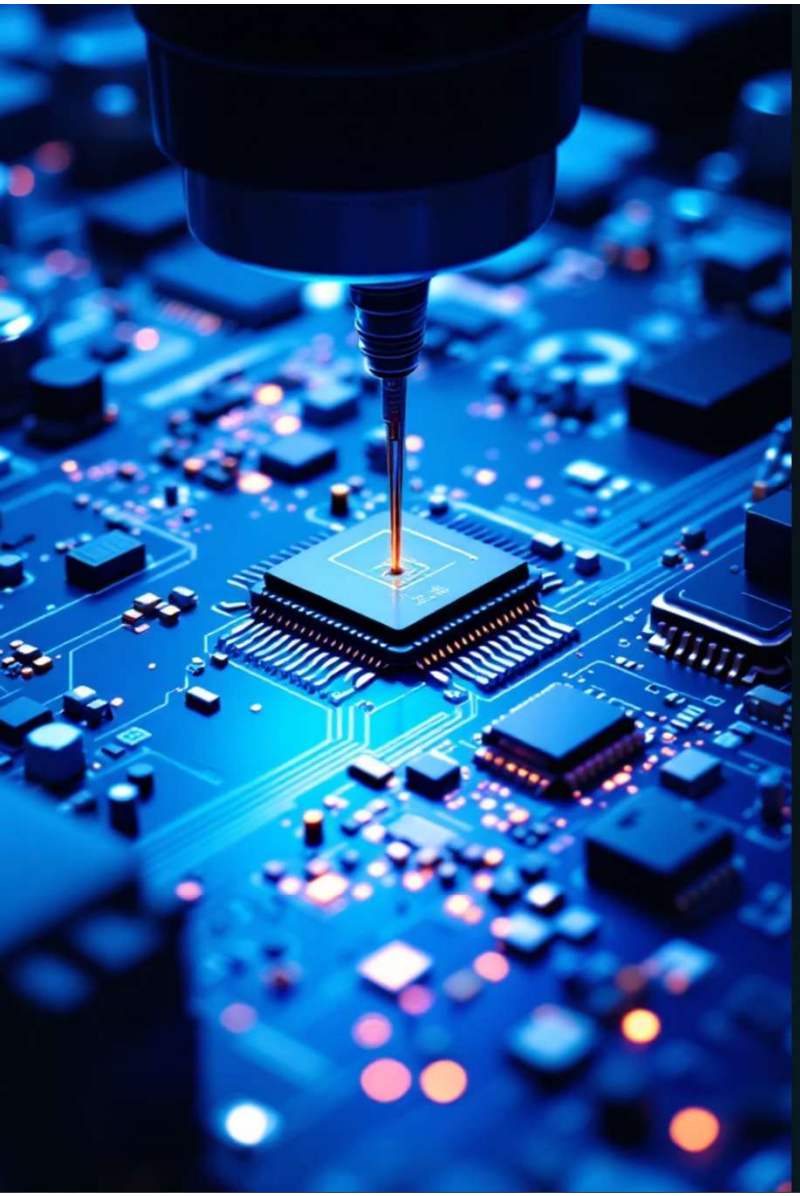


AI-Powered Automated Defect Detection System

Real-Time Quality Inspection Using Deep Learning

Team Name: NULL VECTORS

Team Members: A GAGAN,
T CHIDVILAS BHAGAVAN,
R BHANU PRAKASH,
K PAVANITHA

Hackathon: CONCURRENECE-RIPPLE2K26

Duration: 24 Hours

📌 Problem Statement

Develop a software-based AI/ML model that can automatically detect and classify defects in images of manufactured components.

Real time challenges

Manual Inspection Challenges

- Slow & Expensive
- Inconsistent & Prone to Error
- Human

Targeted Defects

- Cracks
- Detect damaged parts
- Scratches

An accurate, automated, and scalable inspection system is required.

Proposed Solution

We developed an **AI-based automated defect detection system** that:

→ **AI/Machine Learning for Detection**

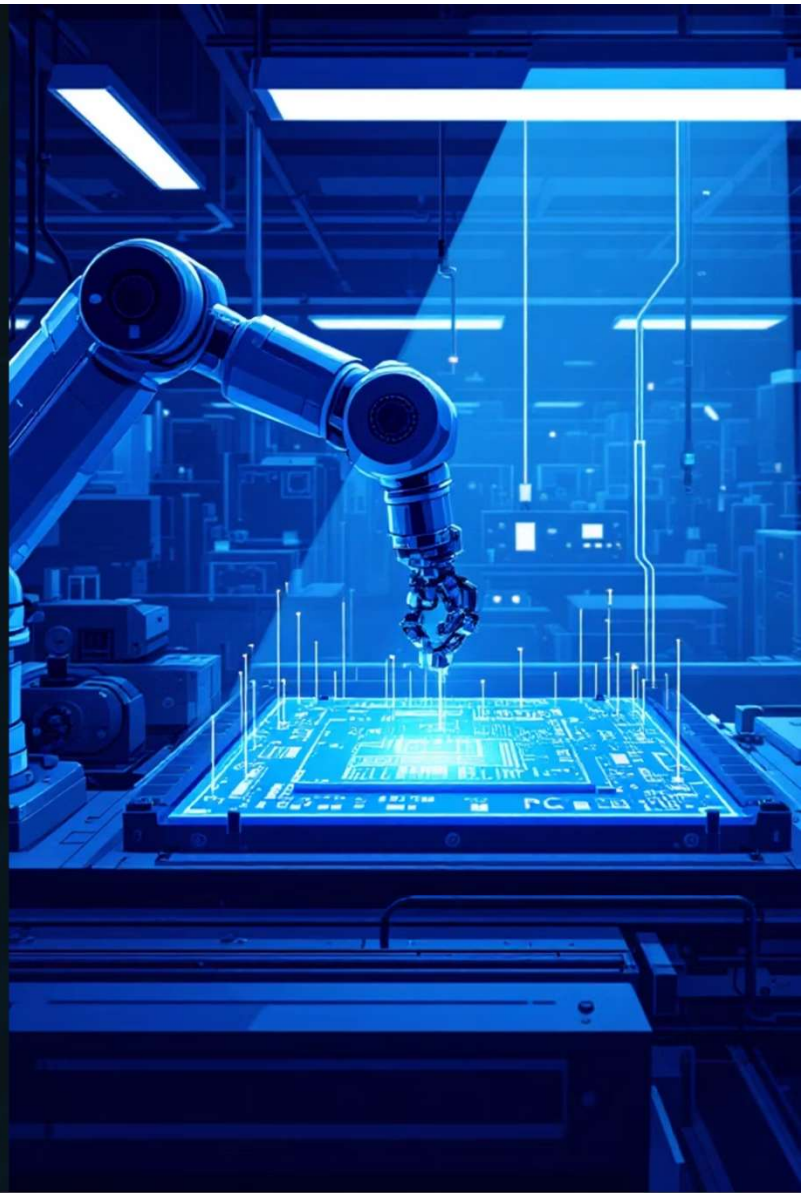
Uses AI/ Machine learning for object detection and real-time inspection.

→ **Instant Output**

Detects, localizes defects, and provides instant PASS/FAIL results.

→ **Industrial Integration**

Can be integrated into industrial production lines for faster, more reliable, and scalable quality control.



Tech Stack Overview

AI & Machine Learning

- TensorFlow
- Py Torch
- Matplotlib

Computer Vision

- OpenCV
- YOLOv8

Deployment

- Web-based Interface

Hardware

- CPU
- Python 3.10+

System Architecture & Workflow

Upload Image

Run YOLOv8

Detect Defects

Evaluate Confidenc e

PASS/FAIL

The system is optimized for real-time industrial use.

Impact's & Benefits

Impact's :

- Real-time defect detection
- Accurate bounding box localization
- Reduced inspection time
- Consistent quality analysis

Industrial Benefits:

- Reduced labor cost
- Faster inspection
- Improved product reliability
- Scalable for smart manufacturing